Answer key. Answers are in magenta.

NRES 746 – Week 5, Lecture 1: Choosing a Data-Generating Model

Instructions: Work alone for about 8-10 minutes, then compare with a neighbor. Several of these have more than one defensible answer, so argue for your choice where you and your neighbor disagree. You do not need to estimate any parameter values.

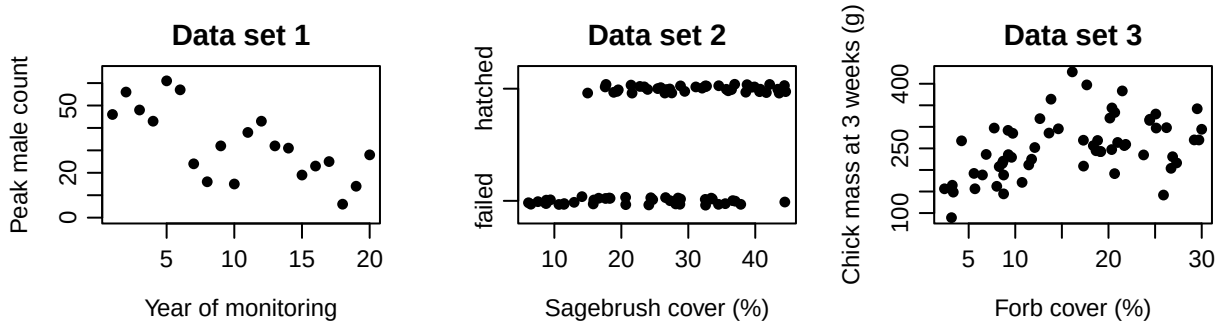
The study system

Suppose a state wildlife agency runs a long-term monitoring program for greater sage-grouse in a Great Basin study area. For each data set below, propose a **data-generating model**: a deterministic function for the *expected value* of the response (which can be as simple as a constant, if nothing predicts it) and a probability distribution for the noise around that expected value. Write each model as a deterministic line and a stochastic line, for example $\mu_i = a + bx_i$ and $Y_i \sim \text{Normal}(\mu_i, \sigma)$, where μ_i is the expected value of Y_i. Write the distribution with its mean set to μ_i where you can, and name any other parameter it needs to describe the spread (like σ for the normal). If you are not sure how to set the mean for your distribution, write "mean $= \mu_i$" anyway and note which parameter controls the spread.

NOTE: the expected value is not necessarily a *parameter* of the stochastic distribution (although it is for the normal and Poisson distributions). Don't worry about that right now- just think of the deterministic function as modeling the expected value and the stochastic distribution as a way to describe the unpredictable variability (spread) in the response.

Here's your grab bag (Bolker Ch. 3 and 4). Use any of these, in any combination:

- **Deterministic functions (for the expected value):** linear, quadratic, power law, negative exponential, Ricker, Michaelis-Menten, logistic
- **Distributions (for the noise):** Bernoulli/binomial, Poisson, negative binomial, uniform, normal, lognormal, gamma, exponential, beta



Part 1: Data set 1 – lek counts

Each spring, biologists count the males displaying at a lek (a traditional breeding display ground). The plot shows the peak count at one lek over 20 years. Counts trend downward at roughly a constant percentage per year, and the scatter around that trend is large.

Expected value: negative exponential Noise distribution: negative binomial

1a. Write your model.

$$\mu_t = N_0 e^{-rt}, \qquad Y_t \sim \text{NegBin}(\text{mean} = \mu_t,\ \text{size} = k)$$

(The data were simulated with $N_0 = 55$, $r = 0.06$, $k = 8$.)

1b. Why this distribution?

Counts are discrete and non-negative (Poisson or negative binomial), and the scatter is too large for a Poisson.

Part 2: Data set 2 – nest fate

Crews monitor 80 nests. Each nest either hatches at least one egg (success) or fails. Crews also measure the percent sagebrush cover around each nest. Nests in denser cover look more likely to succeed. (The expected value of a 0/1 response is the probability of success, and a probability can never go below 0 or above 1, however much cover a nest has.)

Expected value: logistic Noise distribution: Bernoulli

2a. Write your model.

$$p_i = \frac{1}{1 + e^{-(a + b\,\text{cover}_i)}}, \qquad Y_i \sim \text{Bernoulli}(p_i)$$

(Simulated with $a = -2.5$, $b = 0.1$.)

2b. Why this distribution?

The response is a single success/failure (0 or 1) for each nest.

Part 3: Data set 3 – chick mass

Crews weigh chicks at 3 weeks of age and record the percent forb cover at the brood-rearing site (forbs host the insects chicks eat). Mass rises with forb cover, but the gains shrink at high cover. Mass is always positive.

Expected value: Michaelis-Menten Noise distribution: gamma (lognormal also works)

3a. Write your model.

$$\mu_i = \frac{a\,\text{forb}_i}{b + \text{forb}_i}, \qquad Y_i \sim \text{Gamma}(\text{mean} = \mu_i,\ \text{shape} = \alpha)$$

(Simulated with $a = 350$ g, $b = 5\%$, $\alpha = 25$.)

3b. Why this distribution?

Mass is continuous and positive, and its spread grows with the mean, which a gamma with constant shape does.

3c. If time. If your distribution's own parameters are not the mean, work out how to express them using μ_i. (For a gamma with shape α and scale s, the mean is αs. What scale gives a mean of μ_i, and what does the shape do to the noise?)

Scale $s = \mu_i/\alpha$. Then $\text{Var}(Y_i) = \alpha s^2 = \mu_i^2/\alpha$: a larger shape means less noise relative to the mean (coefficient of variation $1/\sqrt{\alpha}$).

Part 4: If time – a fourth data set

Telemetry crews record how far each radio-collared hen moves between consecutive daily locations (meters). Most daily moves are short (a typical hen moves 100-200 m), but a few days involve moves of more than a kilometer. There is no predictor in this data set.

4. What would you use for the expected value, and which distribution for the noise? Does the expected value need to depend on a predictor at all? Explain.

A constant expected value ($\mu_i = \mu$; no predictor needed) with a positive, right-skewed distribution: exponential, gamma, or lognormal.